

Designing Purposeful Computer Graphics Projects using the UN Sustainable Development Goals and Real-World Data

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Abstract

How can we make the teaching of computer graphics (CG) concepts more engaging and purposeful? This case study explores the project-based assessments of a second-year undergraduate CG-focused elective course. Throughout the semester, CG skills and knowledge are built alongside a scaffolded sequence of project activities and collaborative studio practice. The course attracts a diverse cohort, including learners from non-programming disciplines, so studio practice becomes a valuable way to combine creative literacy with emerging computational fluency. The projects are motivated by a learner-selected *United Nations Sustainable Development Goal* and an accompanying statistic, with the graphical and computational challenge to visually convey: (i) *What is the issue?* (ii) *Why does the issue exist?* and (iii) *How do we bring change?* Applying this real-world lens encourages learners to see CG programming not only as a technical skill, but as a purposeful practice with social impact. We present reflections on assessment techniques and challenges, highlighting how purpose-led, data-informed, interdisciplinary contexts can motivate learners, foster creativity, and position CG as a driver for social change.

CCS Concepts

• **Computing methodologies** → **Computer graphics**; • **Social and professional topics** → **Computing education**.

Keywords

Assessment, computer graphics projects, programming, group work, undergraduate

ACM Reference Format:

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1 Introduction

The ACM/IEEE-CS/AAAI Computer Science Curricula 2023 [Kumar et al. 2024] highlights professional dispositions for the area of *Graphics and Interactive Techniques* (GIT), including the importance of being a creative problem-solver and self-directed life-long learner, as well as being an effective communicator that can collaborate within diverse teams. However, few introductory courses typically address these elements. This case study explores an approach that aims to incorporate these skills into an undergraduate introductory CG course. We offer educator perspectives and reflections on the design and implementation of three projects that use the *United Nations Sustainable Development Goals* (SDG) [United Nations 2025]. The 17 SDGs are a global call for humanity to come together and respond to urgent challenges such as climate change, inequality, access to healthcare and education, and the protection of our natural environment.

Despite the opportunity for computing to contribute to society, computing assessments can overlook socially meaningful contexts [Goldweber et al. 2025; Layman et al. 2007], however, our course seeks to encourage learners by locally situating their projects with a real-world statistic related to a chosen SDG. The course was established in 2020 and inspires learners to creatively engage with learning introductory CG programming concepts and techniques in *Processing*¹ [Reas and Fry 2004, 2006] by using data as a tool for storytelling. The project-based activities motivate learners to construct visual narratives around data, through visually dynamic and interactive interfaces that are crafted via computational means.

Three projects incrementally work to build a narrative of: (i) *What is the issue?* (ii) *Why does the issue exist?* and (iii) *How do we bring change?* Throughout the course, learners engage in individual and group projects with peer and instructor feedback — underlining the importance of evaluating GIT-based CG projects with constructive feedback [Kumar et al. 2024], while also supporting a diverse cohort with varying levels of preparedness. Group work balances these differences through peer-to-peer learning and equitable participation, and mirrors the collaborative, interdisciplinary contexts expected in professional practice.

¹<https://processing.org/>

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2 Related Work

Various approaches to CG education exist [Balreira et al. 2017], such as courses that use CG application programming interfaces (APIs) including *DirectX* [Tan 2011], *OpenGL* [Wünsche et al. 2018], and *Vulkan* [Unterguggenberger et al. 2022], as well as API-free approaches [Chen et al. 2018]. Relatively few courses use the *Processing* language [Reas and Fry 2004] although there are reports that it has been used to teach introductory CG [Dodgson and Chalmers 2017; Pellicer et al. 2009; Schweitzer et al. 2010], creative computing [Anderson et al. 2011; Purewal 2013], as well as CS1 programming [Meysenburg 2011], with some educators believing a GIT focus can help capture student interest [Anderson et al. 2011; Pellicer et al. 2009]. A focus on programming through applications of media-based computation has been shown to enhance CS1-based programming education and learner retention [Guzdial 2003, 2013]. Meysenburg [2011] highlights advantages of using *Processing* for learning to program, including its accessibility for capabilities such as graphics, sound, and animation, as well as being Java-based and having a free integrated development environment (IDE) which is available on *Windows*, *macOS*, and *Linux*.

To become competent in CG, learners need to dedicate time to practical exercises [Roudet et al. 2025] and GIT courses often include learner-relatable exercises, such as Schweitzer et al.'s [2010] paint program, 3D anaglyphs, and ray tracing, Meysenburg's [2011] MP3 player and Pellicer et al.'s [2009] "Lunar Lander" game clone. Assessment examples include developing chroma keying, image recovery, and a clone of "Pong" [Meysenburg 2011]. Dodgson and Chalmers' [2017] assignments included an artistic expression, a "Breakout" game clone, and a creative final project — either a 2D game or an interactive artwork. Purewal [2013] used the web-based *Processing.js* variant for their CS0 course, sharing 12 example CG tasks which also culminated in a "Pong" clone. While these examples are relatable, there is an opportunity for assessments situated within socially meaningful contexts [Goldweber et al. 2025; Layman et al. 2007] to motivate learner engagement with CG concepts.

Studio-based learning can focus on the construction of personalised artefacts alongside peer review, and has been used in introductory programming where it was found to be effective with learners developing smartphone applications [Reardon and Tangney 2014]. Molina et al. [2023] reviewed UN SDGs integration into higher education settings, finding examples of lectures, workshops, and seminars. Additionally, Romero Luis et al.'s [2025] GIT-related web development course included teamwork, with projects themed by SDG12 *Responsible consumption and production*. Also in computing education, Sakzad et al. [2024] explored diverging assessments, where all learners undertook the same learning-as-assessment processes, with initial learner-determined actions resulting in different assessment tasks to that of their peers. This practice encourages metacognition and helps create authentic assessments with meaningful learning and feedback opportunities [Billingsley et al. 2025]. Suselo et al. [2017] identified challenges when teaching CG, such as passive learners with limited peer interaction, however, collaborative approaches in computing have been shown to foster accountability, peer learning, and engagement with open-ended

Table 1: Weekly instructor-led topics from the 2025 instance.

Week:	<i>Processing</i> Topics Presented and Demonstrated:
1	Getting started with <i>Processing</i> , Sequence, setup().
2	Variables, draw(), Debug Stubs, Input, Selection.
3	Loops, Nesting, Helper-Funcs, Arrays, .GIF Export.
4	<i>Guest Speaker</i> , Using Images, Applications of Arrays.
5	Importing Libraries, Audio, Sound Effects.
6	Functions, Variable Scope, The Debugger.
7	Code Modularity, Applications of Functions.
8	Classes, Fields, Methods, Objects, Refactoring.

problems [Billingsley et al. 2025], while teamwork also reflects professional practice, where programming is integrated with broader creative and interdisciplinary processes [Kumar et al. 2024].

Drawing from the literature on *Processing*, contextually meaningful tasks, and peer learning, our course design combines *Processing*'s accessible creative programming environment with studio-based SDG-motivated projects. This approach aims to address common challenges in CG education, such as uneven prior experience and passive learning, by situating technical skill-building within purposeful, data-driven narratives and collaborative studio practice.

3 Course Overview

The University of Auckland is a medium-sized urban public university in New Zealand. Our course is a second-year undergraduate elective within the Bachelor of Design, intended for learners with no programming background. It is delivered over a 12-week teaching semester, with one weekly three-hour on-campus studio session. Each studio begins with a short, approximately 30-minute, instructor-led presentation introducing CG programming topics, and sometimes a guest speaker. These presentations include interactive live demonstrations and *Processing*-based CG development techniques relevant to learner progression throughout the course. For the majority of studio time, learners work on practical projects.

Three projects, each four weeks in duration, focus on a learner-chosen SDG and a related real-world statistic, to encourage divergent project outcomes. The complexity expectations for each project align with the topics covered in the weekly studios, with concepts, applications and techniques introduced each week. A high-level view of topic coverage and flow is shown in Table 1. The assessment design deliberately scaffolds individual and group projects so learners can progressively build technical competency while also practising critical reflection and collaboration. Three short controlled practical programming assessments, which do not use the SDG theme, also occur throughout the semester [Hooper et al. 2025]. The coverage of each aligns with the instructor-led presentations and enables a quick check-in on each learner's progress in learning to program. A breakdown of the assessments is shown in Table 2.

The course attracts learners from a range of disciplines, many of whom bring creative or analytical expertise but have limited prior programming experience. Collaborative projects allow these varied strengths to be combined, supporting peer-to-peer learning and shared problem-solving. The group-based structure of *Assignment 3: Interact* mirrors professional contexts where programming

Table 2: Course assessment structure.

	Assessment:	%:	Type:	Weeks:
A1	A Sequence of Primitives	25%	Individual	1–4
A2	Looped States	25%	Individual	5–8
A3	Interact	35%	Group	8–12
	Controlled Coding Tasks	15%	Individual	3, 7, 10

is integrated with wider design, analytical, and communication processes. In this way, group work not only supports learners in developing computational fluency but also prepares them for real-world interdisciplinary collaboration. During the practical project work, work-in-progress critiques and feedback occur, not only with the instructors reviewing each learner's progression, but also with peer review and feedback — encouraging regular weekly progress to be made. An instructor-to-learner ratio of 1-to-25 is generally achieved, and this enables sufficient one-on-one studio-based feedback and critique. Learners are also expected to complete 9.5 hours per week on self-directed project activities outside of studio time, focusing their efforts on furthering their CG project design and development.

3.1 Supporting Technologies — Tools & Services

Beyond *Processing* and its IDE, the course uses two online learner-centric communication venues and services, with learners required to bring their own device (BYOD) to each studio. Critically, the real-time online white-boarding tool *Miro*² is used to collect project artefacts, including SDG and statistic-related research, design process photos, drawings, and sketches, source code snippets, screenshots, and personal commentary notes alongside work-in-progress. *Miro* facilitates group work with its collaborative nature and feedback and ticketing systems, giving learners a venue for constructive critique and a transparent means to hold each other accountable. For each project, an instructor-provisioned template is provided, and learners add artefacts to their board throughout the assessment. Providing a template that models good practice allows students to develop their own working processes, fostering metacognition, and encouraging life-long learning and professional practice. The board is a critical element of the assessment process, as it captures evidence of both the final product and the process undertaken to design and develop the work — as well as peer and instructor critique and the learner's response and subsequent project progression. A free "Education plan" is used, where 100 members can join the course-focused *Miro* team, which enables template sharing, and importantly, has the ability to export the *Miro* as a .PDF at the "Best quality" setting, forming part of the project submission.

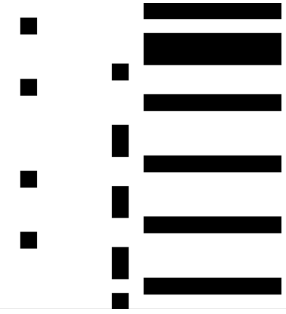
The communications platform *Discord*³ facilitates learner engagement and interaction outside of studios. Learners share their work-in-progress for peer critique, as well as critiquing each other's artefacts. Within a *Discord* server, channels are set up by the instructors for the following course activities: *#announcements*, *#discussions*, *#share-your-work*, and *#group-presentations-feedback*. Throughout each project, learners capture screenshots of the feedback given and received, and add them as evidence to their *Miro*.

²<https://miro.com/>

³<https://discord.com/>



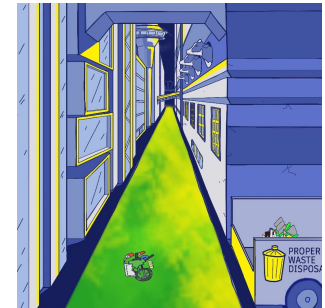
(a) A high-level snippet from the online blog post which shares the artefacts captured during *Assignment 1: A Sequence of Primitives* showing ideation, design, development, personal reflective commentary and final output.



(b) A1: Black squares move horizontally from left to right, and when reaching the middle of the frame, each black square smears towards the right edge — abstractly representing pollution occurring in waterways ("What is the issue?").



(c) A2: An animated loop, with no detectable start or end, where water falls from three spoons that represent: dairy farming, a city, and a factory — i.e., contaminated water spills from each spoon to the environment below ("Why does the issue exist?").



(d) A3: An interactive scene where rubbish in the urban/industrial river can be clicked on using a mouse, and then dragged and dropped in to the disposal truck at the bottom right — the river changes as contamination is cleared ("How do we bring change?").

Figure 1: An example of the three project assessment progression [Hu 2021], inspired by the learner-chosen SDG14 *Life below water* and accompanying statistic: "95–99% of river length in urban, pastoral, and exotic forest areas exceeds water quality guidelines" [Ministry for the Environment 2020].

4 Summative Assignments — The CG Projects

Learners start the course by first reviewing the 17 SDGs [United Nations 2025] and selecting one to explore further. They are tasked with finding a relevant New Zealand-based statistic to situate their chosen SDG, which is then used with each project.

4.1 A1: A Sequence of Primitives (Weight: 25.0%)

The goal of *Assignment 1: A Sequence of Primitives* (A1) is to individually create an animated .GIF that contains computationally-driven geometric primitives that convey "What is the issue?" of the learner's chosen SDG and statistic. Figure 1b shows a static frame from a learner's .GIF that explores SDG14 *Life below water* and the issue of polluted waterways. Initially, the SDG and statistic are recorded within the learner's *Miro* in the *Focus SDG* and *My Statistic* template areas. Learners must provide a short commentary on the



Figure 2: Example selected frames from an *Assignment 1: A Sequence of Primitives* showing the animation's progression, from left-to-right: A new garment with price tag and sparkling stars; garment falls downward; price tag detaches from garment; trash begins to pile up, revealing text describing the statistic; finally, the animation ends with a large trash pile [Chin [n.d.]].

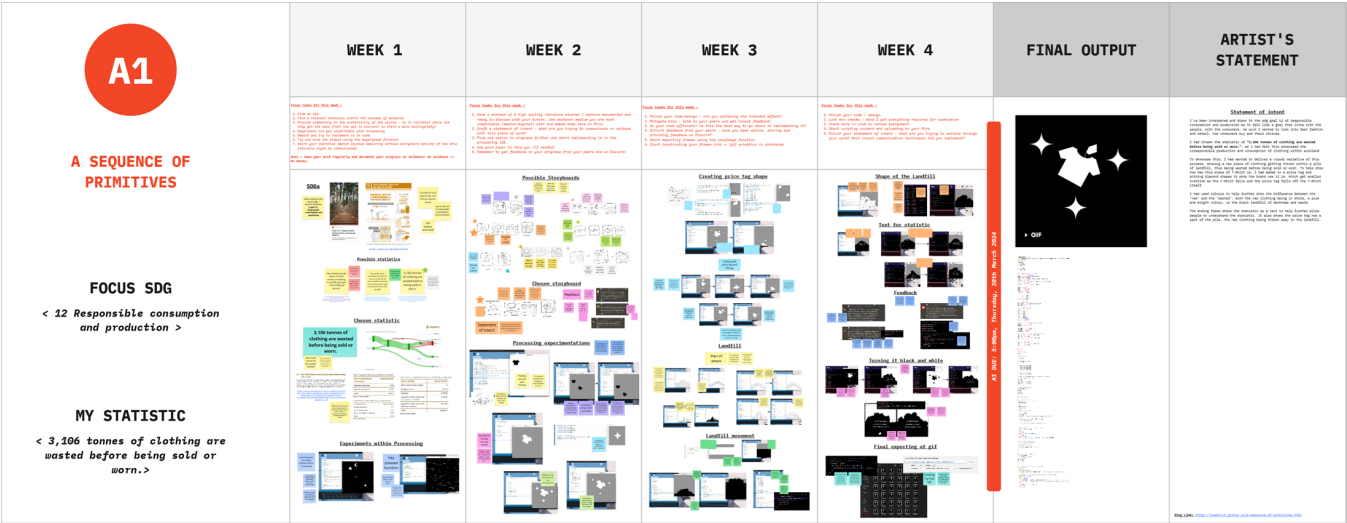


Figure 3: An example high-level view of a *Miro* board from *Assignment 1: A Sequence of Primitives* which displays artefacts captured by a learner throughout the project, including: concept ideation, experimental *Processing* source code and graphical outputs, peer critique and feedback, learner commentary, and final output alongside their artist's statement [Chin 2025].

authenticity of their statistic, and the relevance of the data source. Additional references can also be evidenced to support the validity of the statistic. Learners are encouraged to undertake a process of creative interpretation and design abstraction in creating their project narrative — considering fundamental design principles such as contrast, movement and visual flow, balance and asymmetry, scale and proportion, perspective and depth, and visual rhythm. To ensure they focus on these key principles, a constraint is placed on the A1 project: the geometric primitives used may only be black, white, and grey-scale. Learners apply trigonometry and CG transformation matrices to realise their animations. A sequence of static images from a completed A1 project can be seen in Figure 2.

In the *Miro* template, weekly panels have a list of high-level “Focus Tasks” for learners to follow. A1’s week 1 tasks include creating an initial ideation of how the statistic might be communicated visually. This ideation is expanded in week 2 into six different iterative sketches, each a short storyboard sequence documented on the *Miro* as work-in-progress. These include hand-drawn, digital, or quick *Processing* sketch experiments. Week 2 also requires learners to draft a “Statement of Intent” for what they aim to communicate with their design. A focus task encourages learners to get feedback at this point from their peers, both in-studio and via *Discord*. Learning to program continues to progress each week, with scaffolded

Table 3: Instructor-led topics covered during Assignment 1.

Week:	Topic Coverage — Programming and Processing:
1	setup() {}, size, background, fill, circle, rect, // Commenting, strokeWeight, strokeJoin.
2	draw() {}, rectMode, int, float, boolean, Operators, ellipse, arc, triangle, quad, println, saveFrame, keyPressed() {}, if, else, Relational Operators, translate, scale, rotate, pushMatrix, popMatrix.
3	for, while, Nesting, random, color, frameRate, Math Helper Functions (e.g., sin, abs, constrain), noSmooth, noStroke, Arrays, new and [].
4	PImage, loadImage, image, Logical Operators, .resize, .loadPixels, .pixels[], createImage.

topic introductions, *Processing* demonstrations, and formative exercises — the progression of this during A1 can be seen in Table 1 and the specific *Processing* functionality and topic coverage is shown in Table 3. To complete A1, only the *Processing* topics covered during weeks 1 and 2 are necessary. Week 3 tasks include reminders to seek peer feedback, as well as to test exporting a .GIF using *Processing*’s *Movie Maker* feature. The final tasks from week 4 encourage learners to polish their source code, *Miro*, overall design, and write

Table 4: Instructor-led topics covered during Assignment 2.

Week:	Topic Coverage — Programming and Processing:
5	<code>import processing.sound.*;</code> , <code>SoundFile</code> , <code>.play</code> .
6	Functions, Parameters, <code>return</code> , <code>void</code> , PDE Debugger, Breakpoints, Stepping, Watching, Using <code>sort</code> .
7	More applications of Functions, Debug Tracing.
8	<code>class</code> , <code>new</code> , <code>this</code> , Constructors, Methods, Objects.

a 150–200 word “Artist’s Statement”, as well as to make last-minute checks to ensure they have prepared all expected elements of the submission. As work on the project progresses, each learner’s *Miro* is populated with learner-selected artefacts to document their process, as well as critiques and personal commentary. At the end of the project, the final output is showcased in the form of an animated .GIF and the complete .PDE source code, alongside the learner’s artist’s statement. An additional panel is provided below the weekly panels to collect evidence of feedback given to peers, such as the learner’s *Discord* posts. These artefacts do not directly contribute to the development of the learner’s own project, but demonstrate their efforts to provide constructive feedback on the work of others. Feedback received is expected to be integrated into a learner’s weekly panels alongside reflection and any resulting adjustments made as the project moves forward (e.g., using a virtual *Miro* “sticky note”). An example A1 *Miro* is shown in Figure 3.

4.2 A2: Looped States (Weight: 25.0%)

For *Assignment 2: Looped States* (A2), learners experiment with visual communication strategies, such as colour and form, to create a computational animation which is a seamless loop, without a perceptible beginning or end. In week 5, the focus tasks prompt learners to investigate the contributing and interconnected factors behind their chosen statistic, addressing the question, “Why does the issue exist?” Figure 1c shows a static frame from a learner’s .GIF that posits the cause of polluted waterways is runoff from dairy farms, cities and factories. Learners must consider how the statistic has evolved over time and why it persists in its current form. They employ concepts introduced in class to realise their procedural animation, such as visual effects with alpha-blending, tinting, masking, and linear interpolation. Learners also extend their skills through self-directed exploration of resources such as the *Processing.org* reference documentation, tutorials and *YouTube.com* videos. They apply best-practice design methods, including mood boards, iterative sketching, and planning program components to realise their vision. Art assets can be integrated into the design of this project, and as such, learners may draw elements to be included using software such as *Adobe Illustrator*, *Photoshop* or similar tools. Audio triggers of 1–2 seconds in length are also permissible; however, background audio tracks are not. Learners are expected to create all of their own assets, and not source them from third parties or generative AI (GenAI), to ensure their focus remains on developing their fundamental design and CG programming skills. Table 4 summarises the topics covered during A2, with programming concepts and techniques that support more complex designs, as well as leading towards A3. An example of a static image progression from an A2 project animation loop can be seen in Figure 4.

4.2.1 Marking Criteria (used independently for both A1 and A2):

- *Delivery to format and specification* (5.0 marks): A .ZIP containing the completed *Miro* board as a legible .PDF and .TXT of its live URL; final .PDE source code; animated .GIF.
- *Technical execution and tooling* (5.0 marks): Submission exhibits *Processing* CG programming fundamentals; Appropriate use of software to document, prototype, and iterate on potential solutions — including initiative and self-learning.
- *Critical thinking and qualitative outputs* (10.0 marks): Evidence of an investigation into the cited statistic; Progression from research, iterative development, testing and final output; Commentary alongside work-in-progress; *Miro* presented with care and attention to detail; The final output demonstrates strong aesthetic and visual communication competencies — contextualised, synthesised and rationalised.
- *Active participation and engagement* (5.0 marks): Evidence of in-studio activities; Evidence of online engagement; Sharing work-in-progress and providing critical feedback to peers.

4.3 A3: Interact (Weight: 35.0%)

Assignment 3: Interact (A3) is group-based, and learners select one of their team’s existing statistics to progress further. The project challenges learners to design and develop an interactive *Processing* experience that communicates the question, “How do we bring change?” by showing the user how they can actively influence the chosen statistic. Figure 1d shows an interactive scene that suggests if users remove submerged rubbish from waterways, pollution can be reduced. The team is collectively responsible for delivering the project. Learners are encouraged to explore different input styles, including keyboard and mouse actions (click, press/hold, drag) as well as alternative inputs such as sound and vision, alongside more advanced CG techniques, including curves and particle systems.

The focus tasks within the *Miro* template for A3 (see Figure 5) are purposely left empty, encouraging learners to decide on their own specific weekly tasks and goals. However, the assignment brief recommends that learners conduct precedent research and analysis, examining aesthetic styles, interaction mechanics, interfaces, input devices, and sensory experiences. They are expected to use iterative sketching to storyboard their proposed interaction. Each learner has a dedicated row of weekly panels to document their individual contributions. The template provides a task allocation area, where weekly responsibilities are listed, agreed on by the team, and tracked through to delivery at the next studio. To support accountability, learners provide peer feedback using traffic-light “sticky notes” — green for great work, orange for partial delivery, and red for tasks not completed. The artist’s statement area is expanded to include space for group reflection as well as individual reflections on each team member. Learners are expected to create all project assets, including art and sound. Instructors clarify that the interaction is not intended to be a game, though learners may draw on game mechanics to inform their design.

4.3.1 In-person Group Presentation: The final output is showcased during the week 12 studio, with each team delivering a 5–6 minute presentation, including:

- *Topic:* An introduction of the SDG and chosen statistic, key research insights, and why the statistic is important.



Figure 4: Example selected frames from an *Assignment 2: Looped States* showing a progression of seasons from left-to-right and newly purchased clothing ending up in the trash at the end of the season, with a new garment appearing each season, looped with a yearly seasonal-cycle — the background seasonal images rotate to switch between seasons, and the garment initially rotates in, then drops into the trash can which vertically moves in and out from the bottom of the frame [Chin [n. d.]].

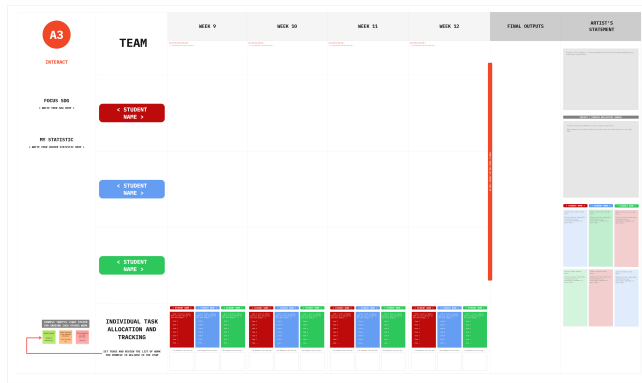


Figure 5: The 2025 Miro template for *Assignment 3: Interact*.

- **Logistics:** Division of tasks and responsibilities, key project milestones, collaboration insights and handover protocols.
- **Process:** Mood board, iterative studies, code development.
- **Narrative:** Design techniques used, art/programming choices.
- **Demonstration:** A live showcase of their *Interact* project.
- **Delivery and Lessons Learnt:** Reflective commentary, specific areas of success, possible improvements, and key decisions.

4.3.2 Individual Postmortem Reflection: A postmortem written reflection has been added to the project, inspired by Hooper et al. [2024]. Learners write a ≈ 1250 -word industry-style reflection, with *Overview*, *What Went Right*, *What Went Wrong*, *Lessons Learnt* and *Conclusion* sections, empowering them to reflect on their processes and outcomes, while also enabling confidential peer critique.

4.3.3 Marking Criteria: The following criteria are used, sharing qualities from Section 4.2.1, but with individual and group mark aspects, and specific *Interact* expectations:

- **Delivery to format and specification (2.5 marks)** [Group]: Also including the group's presentation slides as a .PDF.
- **Technical execution and tooling (7.5 marks)** [Individual].
- **Critical thinking and qualitative outputs (7.5 marks)** [Group].
- **Active participation and engagement (7.5 marks)** [Individual].
- **In-person Presentation (5.0 marks)** [Group]: Meets the target duration; Outlines the team's journey from design to delivery; Includes mood board and inspiration; Includes a live *Interact* demonstration; Concludes with project reflection.
- **Postmortem Reflection (5.0 marks)** [Individual]: A well-written reflection covering the five expected sections, providing insight into successes, shortcomings and lessons learnt.

4.4 Online Showcase — Example Learner Blogs

Earlier versions of the course encouraged learners to share their work via a blog (such as shown in Figure 1a). The following are examples of publicly showcased and online searchable projects:

- SDG2 *Zero hunger* [Boyd 2020]
- SDG3 *Good health and well-being* [Auvele 2020; Huixin 2021]
- SDG5 *Gender equality* [Chen 2020; Huang 2020]
- SDG6 *Clean water and sanitation* [Hou 2022; Taulapapa 2020]
- SDG7 *Affordable and clean energy* [Wang 2020]
- SDG13 *Climate action* [Ly 2021; tmak685 2021]
- SDG15 *Life on land* [Campbell 2022; Thompson 2022]

5 Discussion and Reflections

In our course, learners design purposeful CG projects using SDGs and real-world data as a framework to foster computational fluency and creative literacy. We present our insights into the successes and challenges, alongside learner commentary excerpts from blogs as well as anonymous university-collected feedback (shown in *italics*).

5.1 Engage Learners with Meaningful Contexts

In studio, learners report that merging design, programming, and storytelling creates noteworthy project themes — particularly for those from computing and engineering, where their experience in our course contrasts with their discipline's usual assessments (*"The assessments were fun"*). The choice of SDG and real-world statistic give learners agency to choose a meaningful and motivational reason to engage with the course and results in diverging assessment outcomes (*"Being a Pacific Islander, I wanted to try and fully convey this statistic through coding, in hopes to bring about realization of just how sadistic this is for not only the Pacific Island community in New Zealand, but also the future of the Polynesian community"* [Auvele 2020]). Learning to translate data into compelling visual stories can equip learners with the skills to engage the public in dialogue, raise awareness of critical issues, and contribute to shaping a better world (*"Through this class I have also began (sic) to look at data in a new light"* [Boyd 2020]).

5.2 Scaffold CG Knowledge for Development

Learners' CG programming knowledge and *Processing* skills grow steadily throughout the semester (*"I liked learning a new skill and was able to see my improvement in Processing over the course"*). The complexity of source code increases across the projects, with different applications of mathematics for CG, implementations using classes and objects, containers such as the `ArrayList`, as well as importing modules to expand *Processing*'s functionality. Learners

show evidence of self-directed learning by using a wider range of features than those covered by the instructor (*"I felt like I really took some new knowledge away from this course, which I enjoyed"*). This is inspiring to see, as it shows the course bridges learners into exploring unknown-to-them concepts and CG effects (*"The way the instructor introduced different coding concepts step by step really helped me build a solid foundation, even though I had no prior experience with coding. I appreciated the clear explanations, examples, and the hands-on practice we were given-it made the learning process much more approachable and engaging"*). Several teams have incorporated additional modules in A3, such as *OpenCV* to enable camera input and craft unique interactive experiences, with the assessment made even more engaging by the thought of how their peers will react during the final live demonstrations.

5.3 Capture Iterative Process and Final Product

Miro is central to the course as it documents both process and product, making it a valuable mechanism for assessing authenticity and accountability. Submissions are required in two formats — a .PDF export and a live URL — which together mitigate common challenges. The .PDF ensures that learners who provide links without the correct sharing permissions can still evidence their process, while the live link safeguards against poor .PDF exports or subscription limitations. This dual record also allows instructors to capture the exact state of the *Miro* at submission time, preserving evidence of design, development, critique, and reflection. *Miro* makes patterns of engagement visible: well-populated boards demonstrate steady iteration and peer interaction, while sparse boards reveal when projects have been rushed or when learners have struggled to engage with iterative design and critique. Some challenges remain. To mitigate last-minute rushed artefact creation, future instances of the course will require the submission of more regular .PDF snapshots of a learner's *Miro* to further encourage incremental progress. During the latest A3 team projects, learners identified a limitation with separate rows for individual contributions, which made shared responsibility less visible. Future iterations will trial a shared collaboration panel with ID marker tokens, enabling teams to curate artefacts collectively while still tagging individual contributions.

5.4 CG Teamwork makes the GIT Dream Work

In earlier course iterations, A3 teams of two were formed with fixed leadership roles — one as "Art Director" and the other as "Lead Developer". This often led to siloed contributions, where learners had a tendency to rely on their strengths in either programming or design, rather than contributing equally to their group project. In the most recent iteration of the course, teams were expanded to three members and roles were removed. Each learner was instead designated a "Developer", responsible for both design and programming, fostering shared ownership of source code, art direction, and the overall project. Additionally, the introduction of the individual postmortem has given learners an avenue to provide confidential feedback about their team members, increasing accountability and transparency. For the instructors, this document also reveals previously hidden team collaboration insights, including pain points. Currently, learners share their .PDE files via cloud drives, *Miro*, or *Discord*, which is a significant limitation. Introducing version

control would enhance team collaboration and communication, and mimic professional practice.

5.5 Encourage Peer Engagement with Critique

Learners are positive about using *Discord* to document progress and engage in real-time peer critique and interaction (*"The most helpful was a discord (sic) group discussion section where peers in my class had the opportunity to share with one another and provide help when stuck on certain assignments and progress. Another thing that was helpful was the examples given in class to try on our own"*). However, initial *Discord* engagement is an area for improvement, as some learners are slow to start giving online critiques and peer feedback, despite being reminded during studio sessions. Adding an icebreaker activity at the start of the course, with short formative exercises posted to the *#share-your-work* channel, may encourage earlier project artefact sharing and critique.

5.6 Consider Diverse Disciplinary Backgrounds

Learning to program can be difficult for students from diverse disciplinary backgrounds where this may be their first exposure to CG concepts and techniques and there is a perceived and actual imbalance in programming skills and knowledge (*"I noticed that some classmates who are double majors in design and computer science had a significant advantage when it came to completing the assignments"*). These differences also create opportunities for peer learning, as group projects reflect the collaborative setting in which CG is practised professionally. For future instances, the instructors plan to add a week 1 diagnostic survey to further understand learner backgrounds and aspirations. This may enable them to better connect with learners who struggle, or feel overwhelmed by the programming concepts covered in the course (*"The most challenging part of this course for me was definitely learning how to code, especially since I had no prior experience. At first, even understanding the logic behind basic functions was difficult, and it took a lot of trial and error to make things work. Fortunately, online resources and class examples were helpful, but sometimes I still felt lost during more complex tasks"*).

5.7 Future Directions — Integrating GenAI?

Currently, learners are required to create their own artefacts, keeping the focus of the course on design and CG programming fundamentals. However, there may be positive ways to incorporate GenAI, for example, to assist learners from diverse backgrounds. Image generation could support rapid ideation and iteration, and GenAI could help to develop or explain aspects of source code. Such augmentation could enrich experimentation with programming and design, while ensuring the final outputs remain learner-produced. Additionally, integrating GenAI in other CG course contexts may broaden the application of SDG-framed diverging assessments beyond our course. Moon et al. [2024] present guidelines, for both instructors and learners, to integrate conversational GenAI in creative programming courses. They highlight that their CG beginners could be confused by GenAI responses, that learner abilities were not positively enhanced, and that limiting creativity and over-reliance were risks. Any future moves to integrate GenAI should be carefully considered.

6 Conclusion

Our course shows how learner-selected SDGs can establish diverging assessments, encouraging CG programming skills growth while engaging learners with socially meaningful contexts. This not only motivates learners, but equips them to tell compelling stories and contribute to social change through a CG lens. Learners develop creative problem-solving, communication, and collaboration skills within diverse teams, mirroring the interdisciplinarity of professional practice. In this way, computing can contribute to society by locally situating projects, connecting with learners through a purpose-led, data-informed approach, and we encourage other educators to embed this technique within their own courses.

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